ECE 220 Computer Systems & Programming

Lecture 4 — Programming with the Stack

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Announcements

- **Mock quiz at the CBTF: Sep 7–10** — extra credit, full 25 points just for *participating*
- Use it to try out the testing screens and options (PrairieLearn, the LC-3 web simulator) — so there are **no surprises** during a real quiz
- Register now at `cbtf.illinois.edu/students`
  (each quiz can be scheduled up to 10 days in advance)
- **Midterm 1: Thursday, October 1, 7:00–8:20 pm** — in person, paper, covers Lectures 1–6
  conflict-exam sign-up deadline: Sep 27 (form on the exams page)

Two Notations for Expressions: Infix and Postfix

Infix — what you type

```
7 + (9 - 6) / 3
```

- operator sits *between* its operands
- needs **parentheses** and precedence rules ("/ before +")
- without them the meaning changes: `7 + 9 - 6 / 3` $\neq$ the above

Postfix — operator *after* its operands

```
7 9 6 - 3 / +
```

- each operator applies to the two values written just before it
- **no parentheses — ever**
- unambiguous: one reading, left to right

`9 - 6` $\longrightarrow$ _____

`(9 - 6) / 3` $\longrightarrow$ _____

`7 + (9 - 6) / 3` $\longrightarrow$ _____

Both are worth 8. The question: how do we *evaluate* `7 9 6 - 3 / +`, reading left to right?

Today's Two Questions

Parentheses are just pushes and pops in disguise.

1. How does a stack evaluate *any* arithmetic expression — without parentheses?
2. What must stack programs do when things go wrong — empty pops, full pushes, bad input?

Applications today: palindromes, balanced parentheses, and a working calculator core — the heart of MP2.

Warm-up: Which Stack Is Empty?

Recall our convention: **TOP (R6) points at the top element**; the stack is empty when TOP = STACK_START.

STACK_END	x3FFB	//////	
	x3FFC	x39	
	x3FFD	x0	
	x3FFE	x3	
	x3FFF	x40	
STACK_START	x4000		← TOP

(a)

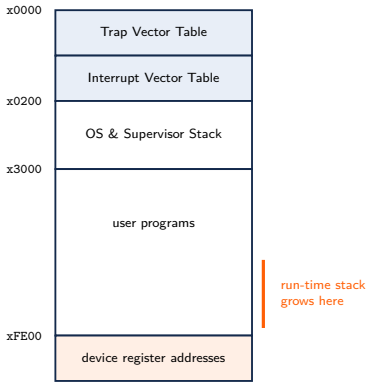
STACK_END	x3FFB	x0	
	x3FFC	x0	
	x3FFD	x0	← TOP
	x3FFE	x0	
	x3FFF	x0	
STACK_START	x4000		

(b)

Which one is empty? _____

The Run-Time Stack

- Each invoked subroutine gets a memory template called an **activation record** (**stack frame**)
- Activation records are **pushed** onto the **run-time stack** in the order the calls happen — and popped in reverse order, exactly LIFO
- This is how every language you will ever use implements function calls
- The **supervisor stack** (OS) is separate — details at the end of the semester



Palindrome Check Using a Stack

A sequence that **reads the same forward and backward**:

`madam` `kayak` *Was it a car or a cat I saw* `123456654321`

How can a stack decide this?

- _____ of the characters onto the stack
- Then, for each character of the second half: _____
- Palindrome $\Leftrightarrow$ every comparison matches

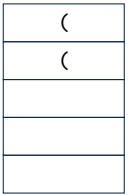
Balanced Parentheses Using a Stack

Balanced: `(()()()())` `((((()))))`
Unbalanced: `(((((((())` `()))` `()()((`

- Open parenthesis `(` : _____ onto the stack
- Close parenthesis `)` : _____ from the stack

Unbalanced is detected two ways:

1. At the end of the expression: _____
2. While scanning: _____



one push per (

Postfix Expressions (Single-Digit Operands)

Infix	Postfix
`(3+4)-5`	`34+5-`
`2^(8-4)`	_____
`7+(9-6)/3`	_____
_____	`512+4*+3-`

Note: `12-` means $1 - 2$, not $2 - 1$.

Are these valid postfix expressions? How would your program know?

- `46*-` _____
- `13+57` _____

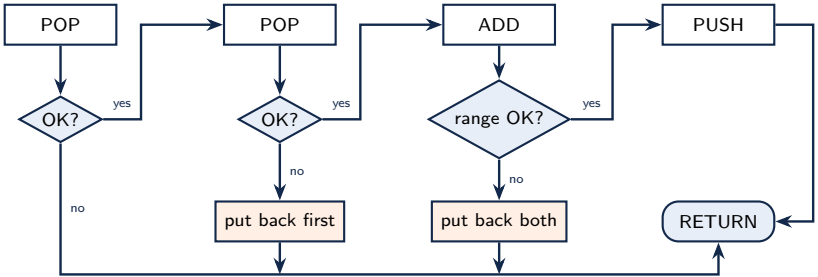
In-Class Trace: Evaluate 34+5−

Token	Action	Stack (top last)
3	push 3	_____
4	push 4	_____
+	pop 4, pop 3, push $3+4$	_____
5	push 5	_____
-	pop 5, pop 7, push $7-5$	_____

- One value left at the end — that is the answer: _____
- For - and /: the **first** pop is the **right** operand

Arithmetic Using a Stack: the ADD Subroutine

Pop two numbers, add them, push the result — *robustly*:



Given (callee-saved): `PUSH` (IN R0; OUT R5), `POP` (OUT R0, R5), `CHECK_RANGE` (IN R0; OUT R5 = 0 iff $-100 \leq$ R0 ≤ 100). **What must ADD watch out for?**

In-Class Problem: Implement ADD

Success path:

```
ADDSUB  ST   R1, A_SAVER1
        ST   R2, A_SAVER2
        ST   R7, A_SAVER7    ; we call subroutines!
; pop first operand; on failure (R5=1) -> A_EXIT

; save it in R1

; pop second; on failure -> A_PUT1

; save it in R2; compute the sum in R0

; check range; on failure -> A_PUT2

; push the sum; then BR to A_EXIT
```

Failure paths (given):

```
A_PUT2  ADD  R0, R2, #0      ; second first...
        JSR  PUSH
A_PUT1  ADD  R0, R1, #0      ; ...then first
        JSR  PUSH
        AND  R5, R5, #0
        ADD  R5, R5, #1      ; report failure
A_EXIT  LD   R1, A_SAVER1
        LD   R2, A_SAVER2
        LD   R7, A_SAVER7
        RET
```

Why push the *second* operand back first?

Note the `ST R7` — ADD calls subroutines, so it must save its own way home.

Summary & What's Next

Today

- Only the **TOP pointer** defines the stack — contents are irrelevant (the warm-up); PUSH/POP from Lecture 3 are today's building blocks
- The **run-time stack** holds one activation record per live call — LIFO is exactly the shape of nested calls
- Stacks solve real problems: palindromes, balanced parentheses, and **postfix evaluation** — push operands; operators pop, compute, push
- Robust stack code checks *every* pop and push, and leaves the stack unchanged on failure

Next lecture

- You now know everything the LC-3 offers: registers, memory, TRAPs, subroutines, the stack. Next: a language that manages them for you — **C**

Code from today: `stack_add.asm` (ADD + POP + PUSH + CHECK_RANGE) in the course code repository.